

PETER STEELE

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petersteele.dev

EMPLOYMENT

Senior Software Engineer	Microsoft - DirectoryStore	February 2024 - Present
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- Design & Implement Backup Checkpoint System, allowing DirectoryStore database system to recover from the loss of the primary replica. Restoring from a backup-checkpoint improves the recovery-point-objective from 24 hours (full resync) to less than 15 minutes (incremental sync).
- Implement automation that reduce livesite toil for the team, preventing ~25 sev2 incidents 25Q1.
- Cross-team collaboration to fix release pipelines, eliminating the root cause for 100+ sev3 incidents 25Q1.

Software Engineer L5	Amazon – AWS Lambda Sandbox Assignment	June 2022 - January 2024
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- Design & Lead automatic NodeLoadShed mitigation project, reducing NodeLoadShed requests by 80% and eliminating manual heat management.
- Build out Lambda Sandbox Assignment Control Plane in five AWS regions
- Implemented control-plane optimizations to improve the speed of data-plane recovery during AZ weigh-in by prioritizing least provisioned partitions

Software Engineer L5	Amazon – Cornerstone Ad Infra	February 2021 - March 2022
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- Cornerstone Ad-Server Lead/Liaison for Thursday Night Football 2021.
 - Collaborated with upstream teams to establish requirements & expectations for Cornerstone team.
 - Scaled ad-servers & load-balancers to support 1.5 million concurrent viewers & cross-region failover.
 - Communicated with developers from services within cornerstone to ensure availability & low-latency.
- Planned & lead scaling efforts for Black Friday / Cyber Monday peak events across ad-server workloads.
 - Zero high-severity incidents during 2021 peak events.
 - Allocated over 100,000 CPU cores to ad-server workloads.
- A few recent interesting work items:
 - Optimized live pipeline templates build time from 4+ hours to <30 minutes, improving productivity.
 - Upgraded ad-server load-balancers to AWS ALB's, resulting in an 8x increase in maximum bandwidth from 4gbps to 33gbps on each endpoint.
 - Migrated ad-programs to next-generation programmable adserver from monolithic adserver.

Software Engineer I / II	Microsoft – Azure Storage	June 2018 - January 2021
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- Scoped, implemented, and deployed an OS image deduplication system for Premium Azure Storage that reduced disk usage by 64PiB, resulting in \$10million/month of premium storage savings.
 - Successful cross-team collaboration with GC team to atomically swap data pointers for zero data loss.
- Mentored intern & guided design direction for “Calculation of Duplicate Data Size” project.

EDUCATION

Blacksburg, VA	Virginia Tech	Fall 2014 - Spring 2018
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- Majors: B.S. in Computer Science, Math, Data Science (CMDA), Economics
- GPA: 3.53

RESEARCH, AWARDS, CONTESTS

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- 1st place at Virginia Tech Undergraduate Research in Computer Science Spring Symposium 2017 (Capstone Award)
 - 2nd place at Virginia Tech Undergraduate Research in Computer Science Spring Symposium 2016 (Faculty Choice)
 - **Two-time International Collegiate Programming Contest (ICPC) World Finalist (2016 & 2017)**

LEADERSHIP

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- President of Programming Team Offseason (Spring 2017/2018): Lectured on a variety of data structures & algorithms topics to 30+ students, attracted Bloomberg as corporate sponsor.
 - Virginia Tech Programming Team Co-coach (Spring 2018): Lead team to qualify for ICPC World Finals 2018
 - Undergraduate Teaching Assistant for CS4104 (Data and Algorithm Analysis) (Spring 2017/2018)

LANGUAGES

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- C++, C#, Java, GoLang, Hack/PHP